

2 **Goodness-of-fit for multi-distribution neutrino** 3 **cross-section measurements with shared events**

4 **F. Martínez López and T. Mohayai**

5 *Department of Physics, Indiana University,*
6 *727 E. Third St., Bloomington, IN, USA, 47405*

7 *E-mail: frmart@iu.edu*

8 **ABSTRACT:** When multiple differential cross-section measurements are extracted from a common
9 event sample, the event-sharing structure between distributions imposes exact linear constraints
10 among the bin counts that reduce the effective dimensionality of the measurement below the total bin
11 count. When the statistical covariance matrix accounts for these inter-distribution correlations, the
12 constraints manifest as a rank deficiency. Systematic uncertainties may formally restore invertibility,
13 but the resulting near-singular directions yield unphysically inflated χ^2 values in global goodness-
14 of-fit tests. We present the range-projected χ^2 , a test statistic that restricts the goodness-of-fit
15 test to the subspace carrying independent statistical information, yielding a χ^2 with $N_{\text{bins}} - N_{\text{null}}$
16 degrees of freedom, where N_{null} is the number of independent constraints. We derive N_{null} from
17 the bin-combination structure of the measurement, and show how it decomposes into structural
18 contributions determined a priori from the binning geometry and kinematic contributions that
19 depend on the phase-space occupancy. The method is validated with an analytical toy model and a
20 realistic simulated neutrino–argon cross-section measurement including unfolding and multi-source
21 systematic uncertainties.

22 **KEYWORDS:** Analysis and statistical methods

23 **ARXIV EPRINT:** [1234.56789](https://arxiv.org/abs/1234.56789)

24 Contents

25	1 Introduction	1
26	2 Constraints from shared events	2
27	2.1 Statistical covariance from shared events	3
28	2.2 Combination matrix and null space	3
29	2.3 Determination of the structural null space	4
30	2.4 Effect of systematic uncertainties	6
31	3 Range-projected χ^2 test statistic	8
32	3.1 Test statistic definition	9
33	3.2 Statistical validity	10
34	3.3 Relation to alternative approaches	10
35	4 Toy example	10
36	4.1 Model and measurement setup	10
37	4.2 Null-space structure	12
38	4.3 Goodness-of-fit validation	13
39	4.4 Parameter estimation	14
40	A Algorithm	15

41 1 Introduction

42 Modern neutrino cross-section literature is experiencing an increase of published measurements
43 reporting multiple differential distributions extracted from the same event sample. MicroBooNE
44 has reported simultaneous measurements of cross sections for final states with and without protons
45 [1, 2], multi-differential QE-like cross section measurements in kinematic imbalance variables
46 [3, 4], and triple-differential inclusive results [5]. Other relevant examples include the multiple
47 final-state kinematic projections from T2K [6, 7], and double-differential inclusive results from
48 NOvA [8–10]. In each of these analyses, the same selected events populate bins in multiple
49 reported distributions, leading to strong inter-measurement statistical correlations. Separate but
50 related efforts have produced simultaneous measurements across different nuclear targets [11–13]
51 and combined neutrino-antineutrino results [14], where correlations arise from shared systematic
52 uncertainties rather than shared events.

53 When multiple distributions are extracted from a common event sample, the event-sharing
54 structure imposes exact linear constraints among the bin counts. If every selected event populates
55 one bin in each distribution, then the total number of events in all of them is exactly identical. More
56 generally, the pattern of shared bin occupancy across distributions determines a set of constraints that

reduce the effective dimensionality of the measurement below the total bin count. These constraints are a property of the measurement itself, independent of any particular statistical treatment.

The blockwise unfolding framework of Ref. [15] provides a rigorous method for computing the full cross-distribution covariance matrices in such measurements. In this approach, all bins corresponding to the same measured observable in the multi-distribution analysis are organized in blocks, each corresponding to a single differential cross-section distribution. Each bin block is unfolded independently using any standard method (e.g. iterative D’Agostini unfolding [16] or Wiener-SVD [17]), while the full covariance matrix, encoding the correlations between bins from different distributions, is constructed and propagated through the cross section extraction procedure. The statistical covariance for bins in different blocks is determined by the number of events they share. The constraints described above can manifest as a rank deficiency in the statistical covariance matrix, as certain directions in bin space carry identically zero variance. The number of these degenerate directions is exactly predictable from the binning geometry.

A natural use of these combined measurements is computing global goodness-of-fit metrics, such as a χ^2 statistic, that test a model against all distributions simultaneously. Compared to evaluating each distribution independently, a global test that accounts for all inter-block correlations provides the maximum discriminating power available from the data. Such global metrics are essential for interaction model tuning efforts [18–20], where measurements of multiple kinematic distributions provide complementary sensitivity to different aspects of the underlying physics. However, a global χ^2 that does not account for the constrained directions yields a test with the wrong effective number of degrees of freedom. When the statistical covariance matrix is constructed under the Poisson assumption, the constraints appear as exact null directions that prevent direct matrix inversion. Systematic uncertainties may formally restore invertibility, but the resulting near-singular directions produce unphysically inflated χ^2 values. In either case, the test is unreliable unless the constrained directions are properly identified and excluded.

We present a method to correctly compute a global goodness-of-fit test statistic for multi-distribution measurements with shared events. The approach is based on identifying the constrained directions from the bin structure of the measurement and projecting the statistic onto the subspace where independent statistical information is available. We refer to the resulting test statistic as the range-projected χ^2 . We demonstrate the method using a pedagogical toy example and connect it to the established statistical literature on hypothesis testing with singular covariance matrices.

2 Constraints from shared events

Consider a neutrino cross-section analysis simultaneously measuring B different flux-integrated differential cross sections, each with N_b true bins. If the measurement is extracted from a common event sample then each selected event will populate exactly one bin in each distribution it belongs to. Recent MicroBooNE analyses [21] have adopted the blockwise unfolding approach presented in Ref. [15] to simultaneous cross-section measurements.

This method provides a systematic framework for reporting the correlations between the different extracted differential cross sections. It provides a general prescription to construct statistical covariance matrices that respect the correlations between bins in different distributions or blocks. The full inter-block covariance matrix constructed in reconstructed space is propagated through

unfolding using a block-diagonal unfolding matrix, obtained as a direct sum of the individual unfolding matrices computed for each block. The result of the combined measurement is a vector of $N_{\text{bins}} = \sum_{b=1}^B N_b$ cross-section values with an $N_{\text{bins}} \times N_{\text{bins}}$ covariance matrix. The total covariance decomposes as $C_{\text{total}} = C_{\text{stat}} + C_{\text{sys}}$, where the statistical covariance component, computed from the shared-event structure, encodes the constraints described in the introduction.

2.1 Statistical covariance from shared events

The core idea behind the treatment of the statistical correlations in the blockwise unfolding approach is the partition of any arbitrary pair of overlapping kinematic bins X and Y into a set of three non-overlapping bins u , v , and w . The events in the intersection $X \cap Y$ are assigned to bin v , while bin u contains the events in the relative complement of Y in X (i.e. $X \setminus Y$), and bin w corresponds to the relative complement of X in Y (i.e. $Y \setminus X$). Denoting the number of predicted events in each bin as n_i , and assuming these follow independent Poisson distributions, the covariance between the counts in bins X and Y is given by [15]:

$$\begin{aligned} \text{Cov}(n_X, n_Y) &= \text{Cov}(n_u + n_v, n_v + n_w) \\ &= \text{Cov}(n_u, n_v) + \text{Cov}(n_u, n_w) + \text{Cov}(n_v, n_v) + \text{Cov}(n_v, n_w) \\ &= \text{Var}(n_v), \end{aligned} \quad (2.1)$$

as by definition bins u , v , and w have no events in common. This implies that the covariance between any two bins can be estimated from the variance of the events they share.

The variance for a set of Monte Carlo (MC) events can be estimated from the sum of their squared weights. This gives us a general prescription to calculate the elements of the statistical covariance matrix as:

$$(C_{\text{stat}})_{ij} = \sum_{e \in (i \cap j)} w_e^2, \quad (2.2)$$

where w_e is the weight associated to event e belonging to the intersection of bins i and j . In the case of measured data the weights are 1, and the covariance element simplifies to the number of events shared between the bins n_{ij} .

After propagation through the block-diagonal unfolding, the statistical covariance matrix in unfolded space inherits the same rank structure, as the unfolding matrices for each individual block are invertible. This rank structure has a direct interpretation in terms of the number of distinct event-sharing patterns across distributions, which we discuss next.

2.2 Combination matrix and null space

The previous discussion shows that the statistical covariance between any two bins is determined solely by the events they share. We now formalize this observation to derive an exact formula for the rank of C_{stat} , and therefore for the number of directions in bin space that carry zero statistical variance.

For each event e , we can define the indicator vector $\phi_e := (\phi_{e,i})_{i=1}^{N_{\text{bins}}}$ such that $\phi_{e,i} = 1$ if $i \in \text{bins}(e)$ and $\phi_{e,i} = 0$ if $i \notin \text{bins}(e)$, where $\text{bins}(e)$ is the set of bins event e populates across all distributions. The indicator vector acts as an identifier of each event in the combined measurement. For example, in a two-block measurement with 3 bins each, an event populating bin 2 of block 1

132 and bin 1 of block 2 has indicator vector $\phi_e = (0, 1, 0, 1, 0, 0)$. The statistical covariance matrix
 133 can then be re-written in terms of indicator vectors as:

$$C_{\text{stat}} = \sum_e w_e^2 \phi_e \phi_e^T. \quad (2.3)$$

134 This outer-product decomposition makes the rank structure of C_{stat} clear. The rank of the
 135 statistical covariance matrix equals the dimension of $\text{span}\{\phi_e\}$, which depends only on the distinct
 136 indicator vectors present, not on their weights. Since all weights are $w_e > 0$ no indicator vector is
 137 removed from the sum, and the column space of C_{stat} is exactly $\text{span}\{\phi_e\}$.

138 We define the bin-combination matrix Ω as the matrix whose rows are the unique indicator
 139 vectors among $\{\phi_e\}$. Each row of Ω represent a distinct pattern of bin occupancy across the blocks.
 140 By construction the row space of Ω equals $\text{span}\{\phi_e\}$, and therefore:

$$\text{rank}(C_{\text{stat}}) = \text{rank}(\Omega). \quad (2.4)$$

141 The number of degenerate directions is then:

$$N_{\text{null}} = N_{\text{bins}} - \text{rank}(\Omega). \quad (2.5)$$

142 Physically, each of the N_{null} directions corresponds to a linear combination of bins along which
 143 the measurement has identically zero variance. Any component of the residual vector along such
 144 a direction is not perceived by the statistical covariance; it is a direction that the measurement has
 145 no statistical power to constrain. Importantly, both Ω and N_{null} are directly computable from the
 146 MC event-to-bin mapping alone, before constructing any covariance matrix, confirming that the
 147 constraints are a property of the measurement geometry rather than the statistical treatment.

148 A vector $\mathbf{x}$ is in the null space of C_{stat} if and only if, for every event e :

$$\sum_{i \in \text{bins}(e)} x_i = 0, \quad (2.6)$$

149 or equivalently $\Omega \mathbf{x} = 0$. Each unique indicator vector acts as a linear constraint on the null-space
 150 vectors, requiring that the components of $\mathbf{x}$ sum to zero over the bins populated by the different
 151 events. The null space directions can be divided into two physically distinct categories: structural
 152 constraints that follow from the block topology alone, and kinematic constraints that depend on
 153 which regions of phase space are populated.

154 2.3 Determination of the structural null space

155 The combination matrix Ω presented previously is constructed from MC events. One determines
 156 which events populate the different bins and collects the unique indicator vectors. However, the
 157 indicator vectors that can appear are constrained by two distinct mechanisms. The first is purely
 158 structural, the geometry of the binning scheme forbids certain bin combinations. The second is
 159 kinematic, correlations between the different variables render some combinations unreachable even
 160 though they are geometrically allowed. These constraints are imposed by the underlying physics
 161 effects, mainly the differential cross section and the detector acceptance. In the following, we show
 162 that the structural constraints can be determined a priori, without generating MC events, from the
 163 bin edges used.

164 The variables used in a measurement are drawn from a set $\{v_1, \dots, v_D\}$. Each block b uses
 165 a subset of these variables, $\text{vars}(b) \subseteq \{v_1, \dots, v_D\}$. Restricting the discussion to combinations of
 166 single- and double-differential measurements, we have $|\text{vars}(b)| \in \{1, 2\}$ for every b . For each
 167 variable v used in a given block b , the binning scheme defines a set of bin edges $E_v^{(b)}$ that partition
 168 the range of v into intervals. In the case of a two-dimensional block, the edges in the binning
 169 variable may be slice-dependent, but nevertheless these can be collected into a single set of edges
 170 per variable across slices. For each variable $v \in \bigcup_b \text{vars}(b)$, we can define the refined edges:

$$\mathcal{E}_v = \bigcup_{b: v \in \text{vars}(b)} E_v^{(b)}, \quad (2.7)$$

171 with $M_v = |\mathcal{E}_v| - 1$ denoting the number of refined bins for variable v . Since $\mathcal{E}_v$ contains all the
 172 bin edges from every block that uses that variable, each refined bin is contained entirely within one
 173 original bin of every such block.

174 The common refinement of all blocks is the Cartesian product of the refined bins across all
 175 variables, denoted by $\mathcal{R}$. Each cell $c \in \mathcal{R}$ specifies a refined bin index for every variable in the
 176 measurement. For every block b , there exists a unique mapping $\beta_b(c)$ between each cell c and an
 177 original bin in said block. The elements of the indicator vector $\phi_c := (\phi_{c,i})_{i=1}^{N_{\text{bins}}}$ associated to cell
 178 c can be written as:

$$\phi_{c,i} = \begin{cases} 1 & \text{if } i = \beta_b(c) \text{ for any block } b, \\ 0 & \text{otherwise.} \end{cases} \quad (2.8)$$

179 If block b does not use variable v then β_b is independent of the refined bin index in v . Therefore,
 180 multiple cells may produce identical indicator vectors.

181 The structural combination matrix Ω_{struct} is defined as the matrix whose rows are the unique
 182 indicator vectors from the cells of the common refinement $\{\phi_c : c \in \mathcal{R}\}$. By construction, every
 183 indicator vector that appears in the Ω matrix obtained from the MC events is also present in Ω_{struct} ,
 184 therefore:

$$\text{rank}(\Omega) \leq \text{rank}(\Omega_{\text{struct}}) \leq N_{\text{bins}}. \quad (2.9)$$

185 This hierarchy motivates a decomposition of the total number of degenerate directions:

$$\begin{aligned} N_{\text{null}} &= N_{\text{bins}} - \text{rank}(\Omega_{\text{struct}}) + \text{rank}(\Omega_{\text{struct}}) - \text{rank}(\Omega) \\ &= N_{\text{null}}^{\text{struct}} + N_{\text{null}}^{\text{kinem}}. \end{aligned} \quad (2.10)$$

186 The first term, $N_{\text{null}}^{\text{struct}}$, counts null directions that arise purely from the binning geometry and is
 187 computable a priori from the bin edges alone. The second term, $N_{\text{null}}^{\text{kinem}}$, accounts for additional null
 188 directions introduced by phase-space restrictions that exclude geometrically allowed combinations
 189 and can only be determined from Monte Carlo. A description of the algorithm used to compute
 190 Ω_{struct} , including an optimization that restricts the enumeration to shared variables, is presented in
 191 Appendix A.

192 The presence of kinematic null directions indicate that certain regions of the combined bin
 193 space are empty, which may signal that the binning extends beyond the kinematic reach of the
 194 measurement or that it is finer than the available statistics can support. In such cases, merging or
 195 removing the offending bins may be preferable to relying on a projection to handle the resulting

constraints. The structural null directions, by contrast are an unavoidable consequence of measuring multiple distributions from shared events and cannot be eliminated by rebinning.

The structure of the null space of Ω_{struct} depends on how the bin edges for the different blocks overlap. In the simplest case, when blocks do not share variables, every combination of one bin per block is allowed. The resulting null space is $(B - 1)$ -dimensional, with a single constraint that requires all block totals to be equal. When blocks share one or more variables, the allowed combinations are restricted and the dimension of the null space can be different from $B - 1$. The number of null directions depends on how shared variable bin edges are aligned between blocks. Aligned slice edges decouple the measurement into different sub-problems, adding additional sum rules and increasing $N_{\text{null}}^{\text{struct}}$. On the other hand, misaligned edges couple adjacent distributions, potentially reducing the number of constraints. In the general case, the interplay between shared variables, bin edge alignment, and block topology makes a closed-form expression impractical, but the computation of $\text{rank}(\Omega_{\text{struct}})$ from the bin edge specifications is straightforward.

The structural and kinematic components of the null space respond differently to systematic uncertainties. Whether these cause the rank deficiency of C_{stat} to persist, partially resolve, or fully resolve in the total covariance C_{total} depends on the nature of the systematic treatment, which we discuss next.

2.4 Effect of systematic uncertainties

Previously, we established that the event-sharing structure imposes $N_{\text{null}} = N_{\text{bins}} - \text{rank}(\Omega)$ exact linear constraints on the bin counts, which manifest as a rank deficiency of C_{stat} . Whether the total covariance matrix inherits the rank deficiency depends on which of these null directions receive support from the systematic uncertainties. Now, we examine the mechanisms by which different systematic treatments do or do not populate the null space of C_{stat} .

A systematic variation s produces a modified vector of expected events $\mathbf{n}_s$, whose departure from the central-value prediction can be expressed in terms of a shift vector:

$$\delta_s = \mathbf{n}_s - \mathbf{n}_{\text{CV}}. \quad (2.11)$$

The different contributions to C_{syst} are constructed from the outer product of these shift vectors. A null direction $\mathbf{x}$ of C_{stat} remains a null direction of C_{total} if and only if $\mathbf{x} \cdot \delta_s = 0$ for every systematic variation in the uncertainty budget. The question is therefore whether the shift vectors have nonzero projections onto the structural and kinematic null spaces.

Consider first the simplest case, where a systematic s acts by reweighting the reconstructed event counts directly. Each event e is assigned a modified weight $w_e^{(s)}$ while preserving its bin assignments. The shift vector can then be written as:

$$\delta_s = \sum_e (w_e^{(s)} - w_e) \phi_e. \quad (2.12)$$

For any null vector $\mathbf{x}$ which satisfies the null-space condition of Eq. (2.6), the projection of $\mathbf{x}$ onto

the shift vector gives:

$$\begin{aligned}
\mathbf{x} \cdot \delta_s &= \sum_e (w_e^{(s)} - w_e) (\phi_e \cdot \mathbf{x}) \\
&= \sum_e (w_e^{(s)} - w_e) \sum_{i=1}^{N_{\text{bins}}} \phi_{e,i} x_i \\
&= 0,
\end{aligned} \tag{2.13}$$

since the inner sum vanishes for every event. Direct reweighting therefore preserves the entire null space, adding no rank to C_{total} . Flux uncertainties, which modify the neutrino energy spectrum through per-event weights, are the classic example. Cross-section model variations and hadronic reinteraction uncertainties, when implemented as per-event reweighting, behave identically.

A qualitatively different situation arises when the systematic uncertainties are propagated through the response matrix rather than changing the event counts per bin directly. In this approach, adopted as the standard prescription in MicroBooNE for neutrino interaction uncertainties [15], a modified response matrix Δ'_s , mapping true bins to reconstructed bins under the given variation, is constructed for each systematic uncertainty. This matrix is obtained either by reweighting the events used to build the nominal response matrix or from an independent MC sample with varied simulation parameters. In this case, the predicted reconstructed counts for systematic variation s are:

$$\mathbf{n}_s = \Delta'_s \boldsymbol{\varphi}_{\text{CV}}, \tag{2.14}$$

where $\boldsymbol{\varphi}_{\text{CV}}$ is the central-value truth prediction. The shift vector is then:

$$\delta_s = (\Delta'_s - \Delta) \boldsymbol{\varphi}_{\text{CV}}. \tag{2.15}$$

Since Δ is block-diagonal, the difference $\Delta'_s - \Delta$ acts independently on each block, projecting the nominal truth vector through block-specific changes in efficiency and migration. The per-event coupling that makes direct reweighting respect all null directions is absent in this prescription. When the weight of an event changes, it affects simultaneously every block it populates; the response-matrix method, in contrast, processes each block independently through its own sub-matrix. The projection of δ_s onto any null vector, whether structural or kinematic, therefore involves differences in how each block responds to the same truth-level prediction, and in general is nonzero. Response-matrix variations thus break both types of null directions, regardless of how Δ'_s is obtained.

A third mechanism applies when systematic uncertainties are evaluated by directly comparing reconstructed event counts from independent MC samples simulated with nominal and varied parameters. In this case, the shift vector takes the form:

$$\delta_s = \sum_{e'} w_{e'}^{(s)} \phi_{e'}^{(s)} - \sum_e w_e \phi_e, \tag{2.16}$$

where the first sum runs over the events in the varied-model sample and the second sum over the nominal sample. Since each event in both samples populates exactly one bin per block, their difference projects to zero onto the structural null directions by an analogous argument to that used for the direct reweighting. Kinematic null directions, however, can be lifted if the variation scatters events into reconstructed-bin combinations absent in the nominal sample. In that

case, new indicator vectors appear that are not rows of Ω , potentially breaking the corresponding constraints. Detector systematic uncertainties, which usually require independent samples with modified simulation parameters that cannot be captured by event reweighting, are the typical example where this prescription is applied.

In practice, whether the eigenvalues of C_{total} along the formerly null directions are exactly zero or merely smaller than the physical eigenvalues, including these directions in a χ^2 test via standard matrix inversion produces inflated values with the wrong effective number of degrees of freedom. A valid goodness-of-fit test must therefore be restricted to the subspace where the measurement carries genuine sensitivity, which is the role of the projection method developed in the following section.

3 Range-projected χ^2 test statistic

The discussion presented in Sec. 2 established that the event-sharing structure of the multi-distribution measurement imposes N_{null} linear constraints, encoded in the null space of the combination matrix Ω , which emerges as a rank deficiency in C_{stat} . A global goodness-of-fit test statistic must therefore be formulated in the subspace orthogonal to these constrained directions, where independent statistical information is available.

The constraints are most naturally identified in reconstructed space, where the combination matrix is constructed. The non-null subspace can be obtained decomposing $\Omega^T \Omega$. This matrix is a square $N_{\text{bins}} \times N_{\text{bins}}$ matrix with the same column space and null space as Ω itself, making it a more compact representation of the same information. We can write its eigendecomposition as:

$$\Omega^T \Omega = V S V^T, \quad (3.1)$$

where $S = \text{diag}(s_1, \dots, s_{N_{\text{bins}}})$ contains the eigenvalues in decreasing order and V is the orthogonal matrix of eigenvectors. Of these eigenvalues, $r = \text{rank}(\Omega)$ are nonzero. Partitioning $V = [V_r \mid \bar{V}_r]$ into the columns associated with the nonzero and zero eigenvalues, the columns of V_r span the subspace of reconstructed bin space that is not constrained by the event-sharing structure. The orthogonal projector onto this subspace is:

$$P_{\text{reco}} = V_r V_r^T. \quad (3.2)$$

Note that P_{reco} is identical to the projector onto the range of C_{stat} as defined in Eq. (2.3), since C_{stat} and $\Omega^T \Omega$ share the same column space by construction. The combination matrix formulation makes explicit the fact that the projector is determined by the measurement geometry.

Cross-section measurements are typically reported in unfolded space, and it is in that space where goodness-of-fit tests are evaluated. The blockwise unfolding matrix U , which is block-diagonal and invertible, maps reconstructed-space vectors onto unfolded space. The non-null basis vectors propagate as $W_r = U V_r$. Since U is not orthogonal in general, the columns of W_r are not orthonormal, and the projector onto their span takes the general form:

$$P_{\text{true}} = W_r (W_r^T W_r)^{-1} W_r^T. \quad (3.3)$$

This projector identifies the subspace of unfolded bin space that carries independent statistical information, with detector response and unfolding effects properly accounted for. Since U is

invertible, W_r has the same rank as V_r and the number of constrained directions is preserved. A test statistic constructed from the projected covariance matrix would have $r = N_{\text{bins}} - N_{\text{null}}$ effective degrees of freedom.

In practice, the eigenvalues of $\Omega^T \Omega$ that are technically zero take small numerical values. The threshold for identifying these can be set relative to the largest eigenvalue (e.g., $s_i/s_1 < 10^{-10}$). In most cases, a clear spectral gap of several orders of magnitude separates the physical eigenvalues from the numerical noise, making the identification unambiguous.

3.1 Test statistic definition

The goodness-of-fit test is evaluated in unfolded space, where the residual vector is defined as:

$$\mathbf{r} = \hat{\boldsymbol{\varphi}} - A_C \boldsymbol{\varphi}_{\text{pred}}, \quad (3.4)$$

where $\hat{\boldsymbol{\varphi}} = U\mathbf{d}$ is the unfolded background-subtracted data, $A_C = U\Delta$ is the additional smearing matrix [17], and $\boldsymbol{\varphi}_{\text{pred}}$ is a truth-level model prediction. The range-projected χ^2 in unfolded space is:

$$\chi_{\text{proj}}^2 = \mathbf{r}^T (P_{\text{true}} V_{\text{total}} P_{\text{true}})^+ \mathbf{r}, \quad (3.5)$$

where $V_{\text{total}} = U C_{\text{total}} U^T$ is the total covariance matrix in unfolded space and $^+$ denotes the Moore–Penrose pseudoinverse.

The pseudoinverse can be avoided entirely by reducing the problem to r dimensions. One can verify that the pseudoinverse of $P_{\text{true}} V_{\text{total}} P_{\text{true}}$ takes the closed form:

$$(P_{\text{true}} V_{\text{total}} P_{\text{true}})^+ = W_r (W_r^T V_{\text{total}} W_r)^{-1} W_r^T. \quad (3.6)$$

Substituting into Eq. (3.5) and defining the reduced residual $\tilde{\mathbf{r}} = W_r^T \mathbf{r}$ and the reduced covariance matrix $\tilde{V}_{\text{total}} = W_r^T V_{\text{total}} W_r$, the test statistic becomes:

$$\chi_{\text{proj}}^2 = \tilde{\mathbf{r}}^T \tilde{V}_{\text{total}}^{-1} \tilde{\mathbf{r}}, \quad (3.7)$$

which involves only a standard inversion of the $r \times r$ matrix $\tilde{V}_{\text{total}}$. Under the null hypothesis, χ_{proj}^2 follows a χ^2 distribution with $r = N_{\text{bins}} - N_{\text{null}}$ degrees of freedom.

An identical construction can be carried out entirely in reconstructed space. Defining the reconstructed-space residual as $\boldsymbol{\delta} = \mathbf{d} - \Delta \boldsymbol{\varphi}_{\text{pred}}$, the corresponding projected χ^2 is given by:

$$\begin{aligned} \chi_{\text{reco}}^2 &= \boldsymbol{\delta}^T (P_{\text{reco}} C_{\text{total}} P_{\text{reco}})^+ \boldsymbol{\delta} \\ &= \boldsymbol{\delta}^T V_r (V_r^T C_{\text{total}} V_r)^{-1} V_r^T \boldsymbol{\delta} \\ &= \tilde{\boldsymbol{\delta}}^T \tilde{C}_{\text{total}}^{-1} \tilde{\boldsymbol{\delta}}, \end{aligned} \quad (3.8)$$

where the reduced form follows directly from the orthonormality of V_r . Without projectors, the two quadratic forms are equivalent, as the unfolding matrix cancels in the unfolded space calculation. This is a well-known result when using the additional smearing matrix to compare model predictions to unfolded data. However, using projectors breaks this equivalence. In this case, the test statistic in unfolded space contains factors of $U^T U$, as can be seen by expressing it in terms of reconstructed-space quantities:

$$\chi_{\text{proj}}^2 = \boldsymbol{\delta}^T U^T U V_r (V_r^T U^T U C_{\text{total}} U^T U V_r)^{-1} V_r^T U^T U \boldsymbol{\delta}, \quad (3.9)$$

which reduces to Eq. (3.8) only when $U^\top U = I$, i.e. when the unfolding matrix is orthogonal. Since unfolding matrices are generically not orthogonal, the two test statistics yield different numerical values for the same data and prediction. The unfolded-space formulation is the appropriate one for testing published cross-section results, as it accounts for effects introduced by the unfolding in the definition of the non-null subspace.

3.2 Statistical validity

Under the null hypothesis, the residual $\mathbf{r}$ is drawn from a multivariate Gaussian distribution with mean zero and covariance V_{total} . The reduced residual $\tilde{\mathbf{r}} = W_r^\top \mathbf{r}$ is a linear transformation of a Gaussian vector, therefore Gaussian itself with mean zero and covariance $\tilde{V}_{\text{total}} = W_r^\top V_{\text{total}} W_r$. Provided that $\tilde{V}_{\text{total}}$ is positive definite, guarantees that $\tilde{\mathbf{r}}^\top \tilde{V}_{\text{total}}^{-1} \tilde{\mathbf{r}} \sim \chi^2(r)$. The positive definiteness of $\tilde{V}_{\text{total}}$ requires V_{total} to have nonzero variance along every direction in the range of the projector P_{true} , which is generically satisfied. This can be verified by checking that all eigenvalues of $\tilde{V}_{\text{total}}$ are strictly positive.

The validity of the range-projected χ^2 can also be understood from the general theory of quadratic forms of correlated normal variables. Theorem 9.2.2 of Ref. [22] establishes that if $\mathbf{Y} \sim \mathcal{N}(0, \Sigma)$ with $\text{rank}(\Sigma) = k$, then $\mathbf{Y}^\top \Sigma^+ \mathbf{Y} \sim \chi^2(k)$ for any choice of generalized inverse Σ^+ . Assuming the null hypothesis, the projected residual $P_{\text{true}} \mathbf{r}$ is normally distributed with covariance $P_{\text{true}} V_{\text{total}} P_{\text{true}}$, which has rank r by construction. The formulation of Eq. (3.5) with the Moore–Penrose pseudoinverse is a specific instance of this general result. The key distinction from purely numerical approaches to handling singular covariance matrices is that the rank is not determined by an eigenvalue threshold applied to the covariance, but is predicted exactly via the combination matrix Ω from the structure of the measurement.

3.3 Relation to alternative approaches

4 Toy example

The formalism developed previously is now illustrated with a toy model that captures the essential features of a multi-distribution neutrino cross-section measurement: a two-dimensional underlying physics process, a simple detector model, and a multi-block binning scheme in which events are shared across distributions. The toy is design to be minimal but non-trivial. A three-block structure produces a rank-deficient statistical covariance matrix, and a Feldman–Cousins study demonstrates the consequences of ignoring or correctly treating the resulting null directions.

4.1 Model and measurement setup

The toy model mimics a muon neutrino charged-current quasi-elastic-like interaction, producing a muon characterized by its momentum p_μ and scattering angle θ_μ . The joint truth-level differential cross section is parametrized as:

$$f(p_\mu, \cos \theta_\mu) = p_\mu^k e^{-k p_\mu / p_0} e^{\kappa \cos \theta_\mu} (1 + \alpha p_\mu \cos \theta_\mu), \quad (4.1)$$

where the four parameters (k, p_0, κ, α) control the shape of the momentum spectrum, the level of forward peaking, and the strength of the momentum–angle correlation, respectively. A central-value (CV) model with $(k, p_0, \kappa, \alpha) = (3.0, 0.2, 2.0, 0.3)$ is used to build the response matrix and

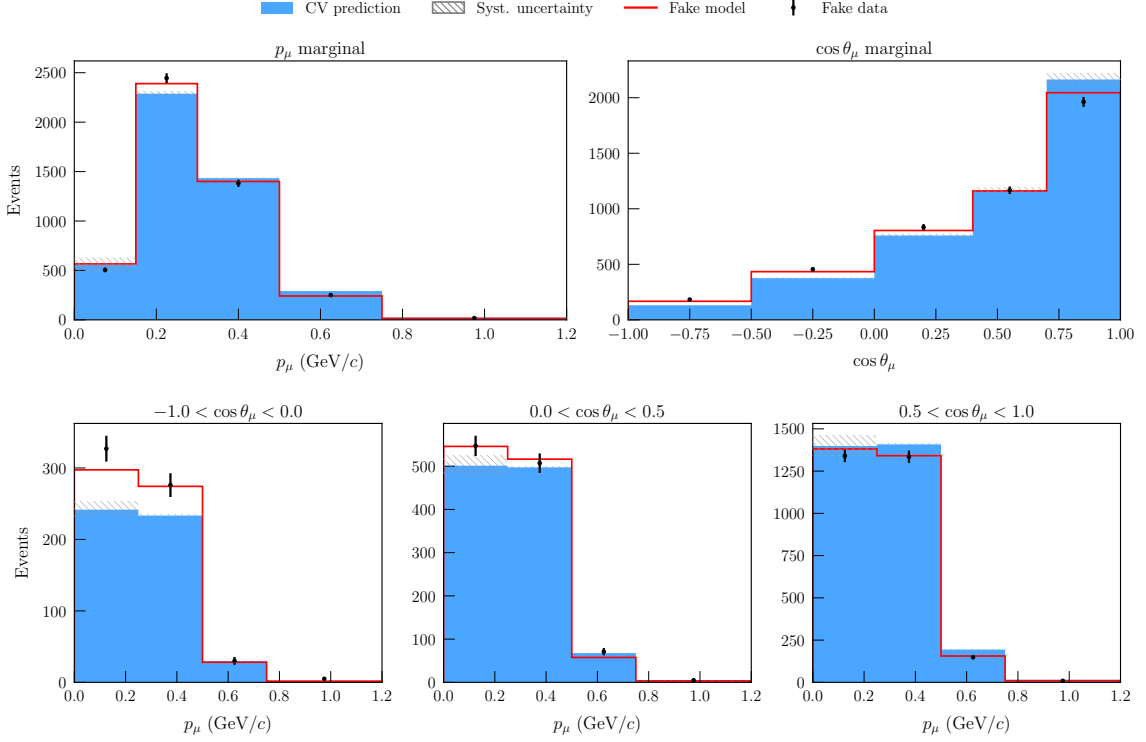


Figure 1. Predicted event distributions in the toy analysis blocks for the CV model (filled histograms) and the fake model (solid lines), together with a pseudo-data realization (points with statistical error bars). The shaded bands indicate the systematic uncertainty on the CV prediction. The top row shows the p_μ and $\cos \theta_\mu$ marginals; the bottom row shows the three $\cos \theta_\mu$ slices of the 2D block, each binned in p_μ .

estimate the covariance, while a “fake” model with $(k, p_0, \kappa, \alpha) = (3.3, 0.2, 1.8, 0.35)$ serves as the true underlying distribution from which pseudo-data are drawn. Truth events are generated via rejection sampling over $p_\mu \in [0.0, 1.2]$ GeV/c and $\cos \theta_\mu \in [-1.0, 1.0]$.

A simple detector model introduces two different effects. Momentum is smeared multiplicatively, $p_{\text{reco}} = p_{\text{true}}(1 + \delta_p)$ with $\delta_p \sim \mathcal{N}(0, \sigma_p)$ and $\sigma_p = 0.05$, while the angle is smeared additively, $\theta_{\text{reco}} = \theta_{\text{true}} + \delta_\theta$ with $\delta_\theta \sim \mathcal{N}(0, \sigma_\theta)$ and $\sigma_\theta = 1^\circ$. A sigmoid efficiency function $\varepsilon(p_{\text{reco}}) = [1 + e^{-(p_{\text{reco}} - p_{\text{thr}})/w}]^{-1}$ with threshold $p_{\text{thr}} = 0.10$ GeV/c and width $w = 0.02$ GeV/c models the turn-on of the event acceptance at low momentum.

The analysis uses three bin blocks in reconstructed space: a p_μ marginal (5 bins), a $\cos \theta_\mu$ marginal (5 bins), and a two-dimensional distribution sliced in $\cos \theta_\mu$ (3 slices, each with 4 bins in p_μ), for a total of $N_{\text{bins}} = 22$. The bin edges of the 2D block do not align exactly with those of the 1D marginals, which is the usual situation in real analyses. A CV sample of $N_{\text{MC}} = 500,000$ truth events is generated, processed through the detector model, and used to construct the response matrix Δ . Pseudo-data samples are generated by drawing $N_{\text{data}} = 5,000$ events from the fake model.

Systematic uncertainties are estimated following the response-matrix approach described in Sec. 2.4. Interaction-model variations are obtained by reweighting the CV sample to six alternative model configurations, each producing a varied response matrix Δ_s . The shift vector $\delta_s = (\Delta_s - \Delta) \varphi$ captures the effect on the predicted reconstructed-space distribution. Detector variations are

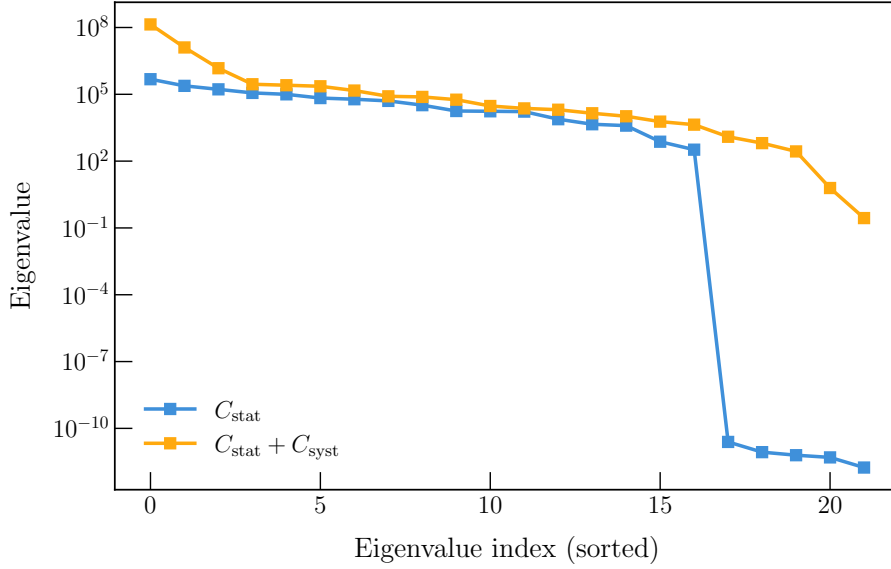


Figure 2. Eigenvalue spectra of the statistical covariance matrix C_{stat} (blue circles) and the total covariance matrix $C_{\text{total}} = C_{\text{stat}} + C_{\text{syst}}$ (orange squares), for the toy analysis, sorted in descending order.

obtained by generating eight independent samples with perturbed detector parameters (σ_p , σ_θ , p_{thr} , and w), each producing a new response matrix from which a shift vector is computed similarly. The total systematic covariance matrix is $C_{\text{syst}} = N_{\text{var}}^{-1} \sum_i \delta_i \delta_i^T$, scaled to data statistics. The statistical covariance is computed using Eq. (2.2), assuming unit event weights, which captures the inter-block correlations from event sharing.

Figure 1 shows the predicted event distributions for the CV and fake models in all three blocks of the toy analysis, together with a pseudo-data sample distribution and the systematic uncertainty bands.

4.2 Null-space structure

The structural combination matrix Ω_{struct} is constructed from the binning configuration of the three blocks, following the prescription from Sec. 2.3. For this bin scheme, Ω_{struct} is rank 17, yielding 5 structural null directions. The combination matrix Ω computed from the MC event list also has rank 17, confirming that all 5 null directions are structural. No additional kinematic constraints arise for this choice of physics model and binning.

The eigenvalue spectrum of C_{stat} confirms this. It shows 5 eigenvalues that are numerically zero, separated from the 17 nonzero eigenvalues by a spectral gap of several orders of magnitude. Adding systematic uncertainties constructed via the response-matrix approach lifts the 5 null eigenvalues, making $C_{\text{total}} = C_{\text{stat}} + C_{\text{syst}}$ formally full rank. However, as shown in Fig. 2, the lifted eigenvalues remain considerably smaller than the bulk of the spectrum. Along these directions the measurement carries no independent statistical information, therefore the variance is determined entirely by the systematic model. Including them in the test statistic introduces poorly calibrated contributions to the χ^2 , as demonstrated next.

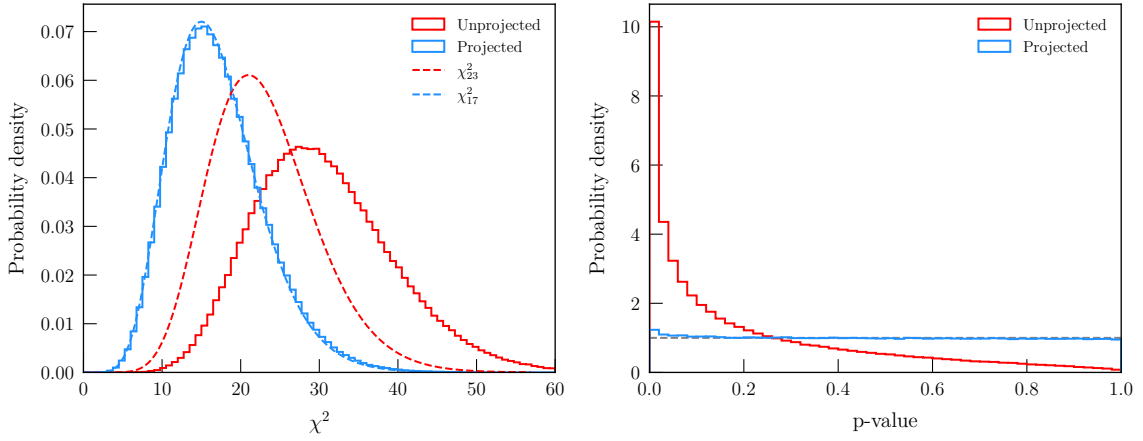


Figure 3. Goodness-of-fit validation for toy experiments generated at the true hypothesis. Left: distributions of the unprojected (red) and range-projected (blue) χ^2 statistics, compared with the expected $\chi^2(22)$ and $\chi^2(17)$ distributions (dashed curves). Right: corresponding p-value distributions, with the false rejection rates at the 5% significance level indicated.

399 4.3 Goodness-of-fit validation

400 The calibration of the projected and unprojected χ^2 test statistics is assessed with a toy Monte
 401 Carlo exercise at the true hypothesis point. A high-statistics simulation of the fake model is used
 402 to estimate the event distribution on a fine reconstructed-space grid, defined by the union of all bin
 403 edges in p_μ and $\cos \theta_\mu$ across the three blocks, corresponding to the common refinement described
 404 in Sec. 2.3. Each toy experiment draws a Poisson-fluctuated total event count and distributes them
 405 across the fine grid via multinomial sampling. These events are then projected onto the three analysis
 406 blocks, preserving the inter-block correlations from event sharing. Systematic fluctuations are added
 407 as correlated Gaussian throws from the Cholesky decomposition of C_{syst} . The χ^2 is evaluated in
 408 reconstructed space, comparing each toy to the forward-folded prediction at the true hypothesis, for
 409 both the unprojected statistic, using the direct inverse of C_{total} , and the range-projected statistic of
 410 Eq. (3.8).

411 Figure 3 shows the results from 10^6 toy experiments. The projected χ^2 statistic closely follows
 412 the expected $\chi^2(17)$ distribution, with a mean of 17.23 and a standard deviation of 6.01, in good
 413 agreement with the expected values of 17 and $\sqrt{2} \times 17 \approx 5.83$. In contrast, the unprojected statistic
 414 is shifted to significantly higher values with a mean of 30.59, and a standard deviation of 8.96, far
 415 above the expectation for a $\chi^2(22)$ distribution of 22 and $\sqrt{2} \times 22 \approx 6.63$. The right panel shows
 416 the corresponding p-value distributions. The projected statistic yields an approximately uniform
 417 distribution, with a false rejection rate of 5.7% at the 5% significance level. The unprojected statistic
 418 produces a false rejection rate of 32.4%, more than six times the expected value. An analyzer using
 419 the unprojected statistic for goodness-of-fit testing would falsely reject the correct model roughly
 420 one third of the time.

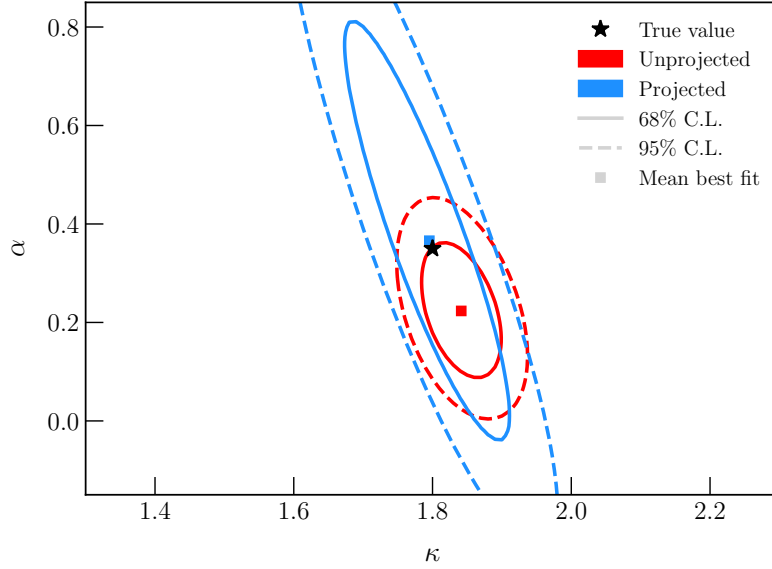


Figure 4. Confidence regions in the (κ, α) plane from the Asimov dataset, showing the 68% and 95% CL contours for the range-projected (blue) and unprojected (red) test statistics. The star marks the true parameter values; the squares indicate the mean best-fit points across 10^6 toys.

4.4 Parameter estimation

The impact of using the projected test statistic on parameter estimation is assessed with a two-dimensional Feldman–Cousins study. A grid of hypotheses is constructed over (κ, α) with $k = 3.3$ held constant. At each grid point, the truth-level prediction is forward-folded through the nominal response matrix and compared to the toy pseudo-data. The test statistic $\Delta\chi^2 = \chi^2(\kappa, \alpha) - \chi^2_{\min}$ is computed for both the unprojected and projected cases, and confidence regions are constructed from the resulting $\Delta\chi^2$ surfaces.

Figure 4 shows the 68% and 95% confidence level contours from the Asimov dataset for both test statistics. The projected contours are correctly centered on the true parameter values, while the unprojected contours are shifted, with the 68% region excluding the true point. The unprojected best fit values are biased, particularly in α . The mean best-fit value across toys is $\hat{\alpha} = 0.22$, far from the true value of 0.35, while the projected statistic recovers $\hat{\alpha} = 0.37$. The bias in κ is milder ($\hat{\kappa} = 1.84$ for unprojected, $\hat{\kappa} = 1.80$ for projected, compared to the true 1.80). This indicates that α has greater overlap with the null directions.

The frequentist coverage confirms these observations. At the 68.3% confidence level, the projected statistic achieves nominal coverage in both the joint two-dimensional region (68.2%) and the one-dimensional profiled intervals (68.1% for κ , 67.9% for α). The unprojected statistic suffers severe undercoverage: 36.5% in the joint region, 44.6% for κ , and 34.1% for α . The undercoverage is worst for α , consistent with its stronger sensitivity to the null-space contamination. The projected contours are larger than the unprojected ones, reflecting the genuine sensitivity of the measurement once the ill-constrained directions are properly removed.

Acknowledgments

A Algorithm

The naive counting of the common refinement $\mathcal{R}$ has cost $\prod_v M_v$, which grows exponentially with the number of kinematic variables. A typical multi-distribution measurement involves $\mathcal{O}(5)$ variables each with $\mathcal{O}(10)$ refined bins, giving $\mathcal{O}(10^5)$ cells. The key insight is that only shared variables (i.e. the ones appearing in two or more blocks) generate the cross-block constraints. Dividing the variable set into shared (V_s) and private (V_p) variables, one has to enumerate only the reduced refinement $\prod_{v \in V_s} M_v$. For each of the shared-variable cells, the bins from blocks that depend on the shared variable are determined by lookup, while blocks whose binning variable is private contribute their bins independently via a local Cartesian product. If the number of shared variables is small the computational cost is dramatically reduced.

References

- [1] MICROBooNE collaboration, *First Simultaneous Measurement of Differential Muon-Neutrino Charged-Current Cross Sections on Argon for Final States with and Without Protons Using MicroBooNE Data*, *Phys. Rev. Lett.* **133** (2024) 041801 [[2402.19281](#)].
- [2] MICROBooNE collaboration, *Inclusive cross section measurements in final states with and without protons for charged-current ν_μ -Ar scattering in MicroBooNE*, *Phys. Rev. D* **110** (2024) 013006 [[2402.19216](#)].
- [3] MICROBooNE collaboration, *First Double-Differential Measurement of Kinematic Imbalance in Neutrino Interactions with the MicroBooNE Detector*, *Phys. Rev. Lett.* **131** (2023) 101802 [[2301.03706](#)].
- [4] MICROBooNE collaboration, *Multidifferential cross section measurements of $\nu\mu$ -argon quasielasticlike reactions with the MicroBooNE detector*, *Phys. Rev. D* **108** (2023) 053002 [[2301.03700](#)].
- [5] MICROBooNE collaboration, *Measurement of three-dimensional inclusive muon-neutrino charged-current cross sections on argon with the MicroBooNE detector*, *Phys. Lett. B* **870** (2025) 139939 [[2307.06413](#)].
- [6] T2K collaboration, *Characterization of nuclear effects in muon-neutrino scattering on hydrocarbon with a measurement of final-state kinematics and correlations in charged-current pionless interactions at T2K*, *Phys. Rev. D* **98** (2018) 032003 [[1802.05078](#)].
- [7] T2K collaboration, *First T2K measurement of transverse kinematic imbalance in the muon-neutrino charged-current single- π^+ production channel containing at least one proton*, *Phys. Rev. D* **103** (2021) 112009 [[2102.03346](#)].
- [8] NOvA collaboration, *Measurement of the double-differential muon-neutrino charged-current inclusive cross section in the NOvA near detector*, *Phys. Rev. D* **107** (2023) 052011 [[2109.12220](#)].
- [9] NOvA collaboration, *Measurement of $d^2\sigma/d|\vec{q}|dE_{avail}$ in charged current ν_μ -nucleus interactions at $\langle E_\nu \rangle = 1.86$ GeV using the NOvA Near Detector*, *Phys. Rev. D* **111** (2025) 052009 [[2410.05526](#)].
- [10] NOvA collaboration, “Measurement of the double-differential cross section of muon-neutrino charged-current interactions with low hadronic energy in the NOvA Near Detector.” 10, 2024.

- [11] T2K collaboration, *Simultaneous measurement of the muon neutrino charged-current cross section on oxygen and carbon without pions in the final state at T2K*, *Phys. Rev. D* **101** (2020) 112004 [2004.05434].
- [12] MINERvA collaboration, *Simultaneous Measurement of ν_μ Quasielasticlike Cross Sections on CH, C, H₂O, Fe, and Pb as a Function of Muon Kinematics at MINERvA*, *Phys. Rev. Lett.* **130** (2023) 161801 [2301.02272].
- [13] MINERvA collaboration, *Simultaneous Measurement of Muon Neutrino ν_μ Charged-Current Single π^+ Production in CH, C, H₂O, Fe, and Pb Targets in MINERvA*, *Phys. Rev. Lett.* **131** (2023) 011801 [2209.07852].
- [14] T2K collaboration, *First combined measurement of the muon neutrino and antineutrino charged-current cross section without pions in the final state at T2K*, *Phys. Rev. D* **101** (2020) 112001 [2002.09323].
- [15] S. Gardiner, “Mathematical methods for neutrino cross-section extraction.” 2024.
- [16] G. D’Agostini, *A Multidimensional unfolding method based on Bayes’ theorem*, *Nucl. Instrum. Meth. A* **362** (1995) 487.
- [17] W. Tang, X. Li, X. Qian, H. Wei and C. Zhang, *Data Unfolding with Wiener-SVD Method*, *JINST* **12** (2017) P10002 [1705.03568].
- [18] C. Wilkinson et al., *Testing charged current quasi-elastic and multinucleon interaction models in the NEUT neutrino interaction generator with published datasets from the MiniBooNE and MINERvA experiments*, *Phys. Rev. D* **93** (2016) 072010 [1601.05592].
- [19] MINERvA collaboration, *Tuning the GENIE Pion Production Model with MINERvA Data*, *Phys. Rev. D* **100** (2019) 072005 [1903.01558].
- [20] GENIE collaboration, *Neutrino-nucleus $CC0\pi$ cross-section tuning in GENIE v3*, *Phys. Rev. D* **106** (2022) 112001 [2206.11050].
- [21] MicroBooNE collaboration, *Measurement of single- and double-differential cross sections for mesonless charged-current muon neutrino interactions on argon with final-state protons using the MicroBooNE detector*, *Phys. Rev. D* **112** (2025) 112004 [2403.19574].
- [22] C. Rao and S. Mitra, *Generalized Inverse of Matrices and Its Applications*, Probability and Statistics Series, Wiley (1971).